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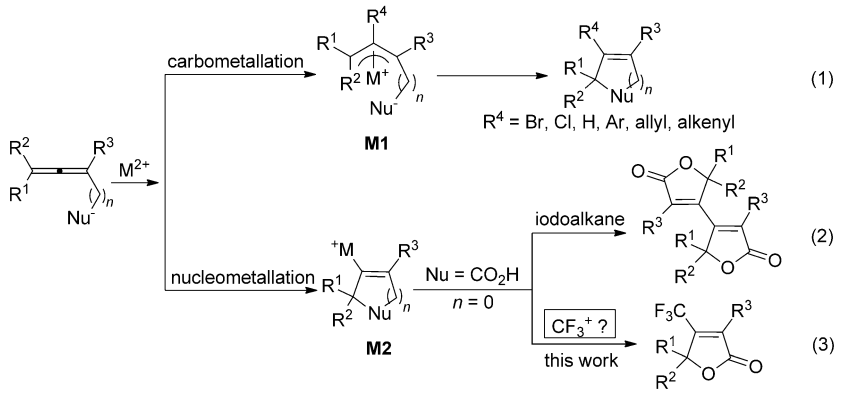
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Copper-Catalyzed Cyclic Oxytrifluoromethylation of 2,3-Allenoic Acids to Trifluoromethylated Butenolides

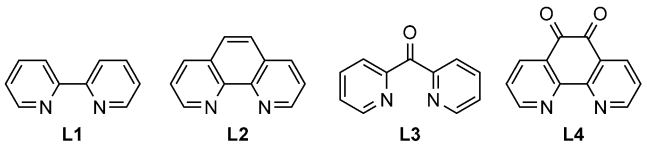
Qiong Yu[a] and Shengming Ma*[a, b]

In recent years, cyclizations of functionalized allenes have been developed to be one of the most efficient methodologies for the synthesis of cyclic compounds via intermediacy of π-allylic metal intermediate **M1** formed by carbometallation or cycloalkenyl metal intermediate **M2** formed by the nucleometalation.[1] We have attempted to synthesize β-methyl butenolides by quenching this cyclic intermediate **M2** with iodomethane (methyl cation). However, the desired product was not formed and the oxidative dimeric cyclization of 2,3-allenoic acids was observed unexpectedly,[2] indicating that the quenching of this type of intermediate **M2** with an electrophile is quite challenging. On the other hand, we reasoned that due to the prevalent high potential biological activity of trifluoromethyl substituted compounds[3] and butenolides,[4] trifluoromethylation of butenolides will be of great interest. As a continuation of our research on demonstrating the new reactivity of functionalized allenes, we proposed to use CF_3^+,[5] to trap intermediate **M2** with the involvement of higher valent metal catalyst, such as Pd^{IV}.[5a–c,6] Herein, we report a copper-catalyzed cyclic oxytrifluoromethylation of 2,3-allenoic acids using a hypervalent trifluoromethyl iodonium reagent, which allows for the synthesis of β-trifluoromethyl[7] butenolides under mild conditions [Eq. (3), Scheme 1].



Scheme 1. Metal-mediated synthesis of functionalized butenolides from 2,3-allenoic acids.

With this notion in mind, we began our studies using Togni's reagents[8] as the electrophiles. When the reaction was carried out with 4-phenyl-2,3-butadienoic acid (**1a**) and Togni's reagent II in the presence of 10 mol% $PdCl_2$ in DMA at 70°C, no trifluoromethylation product was observed [Eq. (1), Scheme 2]. To our delight, when we switched the catalyst system to 15 mol% CuCl with 20 mol% 2,2′-bipyridyl (**L1**), the reaction proceeded to give the desired product **2a**, albeit in 38% yield. When Togni's reagent I was employed, **2a** was formed in 17% yield [Eq. (2), Scheme 2].



To further optimize the reaction conditions, we first tested other copper complexes, such as CuBr, CuI, $CuCl_2$, $Cu(OAc)_2$, and $[Cu(CH_3CN)_4]PF_6$ (entries 1–7, Table 1). It is observed that **1a** could be successfully converted to trifluoromethylated butenolide (**2a**) in the presence of a Cu^I or Cu^{II} complex. CuBr, an easily accessible and cheap Cu^I salt, gave the best results, affording **2a** in 59% yield. A control experiment showed that no product was obtained in the absence of copper complex (entry 12, Table 1). Subsequently, We investigated the ligand effect by applying a series of bidentate ligands: when we used 1,10-phenanthroline (**L2**), the yield was 52% (entry 8, Table 1); the use of di-2-pyridyl

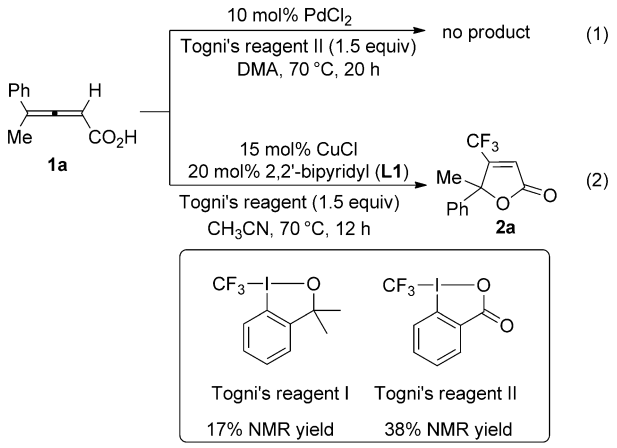
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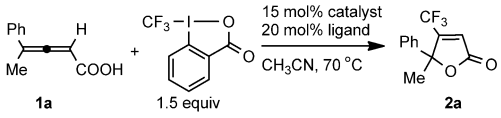
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Scheme 2. Initial trials on the trifluoromethylation reaction of **1a**.

ketone (**L3**) afforded the product in 63% yield; finally, we were glad to observe that by applying 1,10-phenanthroline-5,6-dione (**L4**) the yield of **2a** was further improved to 66% (entry 10, Table 1). In the absence of ligand, a significantly diminished yield was observed (entry 11, Table 1).

Table 1. Effects of catalysts and ligands on the reaction.[a]



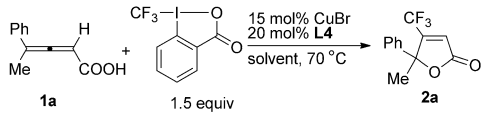
Entry	Copper salt	Ligand	t [h]	Yield of **2a** [%][b]
1	CuCl	**L1**	12	38
2	CuBr	**L1**	11	59
3	CuI	**L1**	11	44
4	$[Cu(CH_3CN)_4]PF_6$	**L1**	11	51
5	$CuCl_2$	**L1**	11	35
6	$Cu(OAc)_2$	**L1**	12	56
7	$Cu(OTf)_2$	**L1**	12	39
8	CuBr	**L2**	12	52
9	CuBr	**L3**	12	63
10	CuBr	**L4**	12	66
11	CuBr	–	12	34
12[c]	–	**L4**	24	0

[a] Reaction conditions: unless otherwise specified, the reaction was carried out by using **1a** (0.2 mmol), Togni's reagent II (0.3 mmol), copper catalyst (0.03 mmol), and ligand (0.04 mmol) in 2 mL of CH_3CN under argon atmosphere. [b] Determined by ^{19}F NMR spectroscopy by using $PhCF_3$ as an internal standard. [c] Recovery 31%.

To further improve the yield, various solvents were surveyed (Table 2). As summarized in Table 2, the reaction could proceed smoothly to afford **2a** in most of tested solvents except toluene (entry 4, Table 2). We were pleased to observe that the product **2a** in chlorine-containing solvents gave decent yields (entries 7-10, Table 2), with dichloromethane being the best, affording **2a** in 92% yield (entry 10, Table 2).

With these optimized conditions in hand, we next explored the substrate scope of the reaction and found that

Table 2. Solvent effect on the reaction.[a]

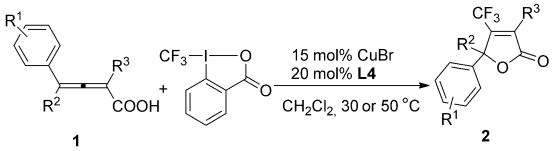


Entry	Solvent	t [h]	Yield of **2a** [%][b]
1	CH_3CN	12	66
2	MeOH	24	62
3	EtOH	25	46
4	toluene	24	0[c]
5	DMF	12	53
6	dioxane	25	46
7	CCl_4	12	52
8	DCE	12	68
9	$CHCl_3$	12	77
10[d]	**CH_2Cl_2**	**12**	**92**

[a] Reaction conditions: unless otherwise specified, the reaction was carried out by using **1a** (0.2 mmol), Togni's reagent II (0.3 mmol), and CuBr (0.03 mmol) and **L4** (0.04 mmol) in 2 mL of solvent under argon atmosphere. [b] Determined by ^{19}F NMR spectroscopy by using $PhCF_3$ as an internal standard. [c] ^{19}F NMR spectrum was complicated. [d] At 30 °C.

various 2,3-allenoic acids can be transformed into desired products in moderate to good yields (Table 3). When R^1 was H, a series of substrates with proton, alkyl and phenyl as R^2 may be applied to afford the corresponding products in moderate to good yields (entries 1–4, 11 and 12, Table 3). Notably, R^1 could also be proton or electron-donating groups (entries 5 and 6, Table 3) or electron-withdrawing groups (entries 7–10, Table 3), giving the desired products in good yields. Such aryl substituents provide an opportunity

Table 3. Copper-catalyzed trifluoromethylation of 2,3-allenoic acids.[a]



Entry	R^1, R^2, R^3	t [h]	Yield of **2** [%][b]
1	H, Me, H (**1a**)	6	89 (**2a**)
2[d]	H, Et, H (**1b**)	9	80 (87)[c] (**2b**)
3	H, n-C_4H_9, H (**1c**)	16	68 (**2c**)[g]
4	H, Ph, H (**1d**)	23	81 (**2d**)
5	4-Me, Me, H (**1e**)	4	81 (**2e**)
6	4-MeO, Me, H (**1f**)	5	84 (**2f**)
7	4-Br, Me, H (**1g**)	23	85 (**2g**)
8[d]	4-F, Me, H (**1h**)	19	85 (**2h**)
9[d,e]	3-Br, Me, H (**1i**)	17	67 (**2i**)
10[d]	3,5-$(CF_3)_2$, Me, H (**1j**)	9	84 (**2j**)
11[d]	H, H, Me (**1k**)	23	60 (**2k**)
12[d,f]	H, H, n-C_4H_9 (**1l**)	24	43 (**2l**)[h]

[a] Reaction conditions: unless otherwise specified, the reaction was carried out by using **1** (1 mmol), Togni's II (1.5 mmol), and CuBr (0.15 mmol) and **L4** (0.2 mmol) in 5 mL of CH_2Cl_2 under argon atmosphere. [b] Isolated yield. [c] The reaction was carried out by using **1b** (6 mmol, 1.1284 g), Togni's II (9 mol) and CuBr (0.9 mmol) and **L4** (1.2 mmol) in 30 mL of CH_2Cl_2 for 17 h. [d] At 50 °C. [e] The reaction was carried out by using CuBr (0.20 mmol, 20 mol%) and **L4** (0.25 mmol, 25 mol%). [f] The reaction was carried out CuBr (0.2 mmol, 20 mol%) and **L4** (0.1 mmol, 10 mol%). [g] 94% purity. [h] 91% purity.

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for further synthetic elaboration. It should be noted that when 4-F or 3,5-$(CF_3)_2$ was employed as R^1 (**1h** or **1j**), this trifluoromethylation reaction also proceeded well (entries 8 and 10, Table 3). H, Me and *n*-C_4H_9 were all competent substituent for R^3. The structure of the product was further confirmed by X-ray diffraction studies of **2g** (Figure 1).[9] When 1.13 g of **1b** was used, the corresponding trifluormethylated butenolide (**2b**) was obtained in 87% yield, representing the robustness of this method (entry 2, Table 3).

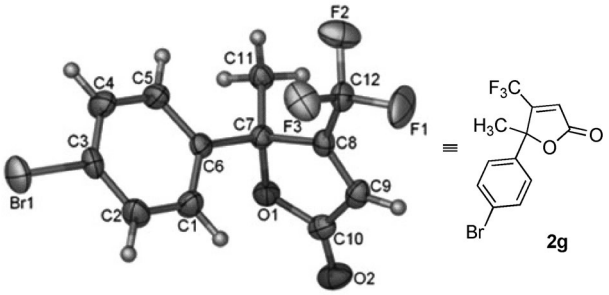
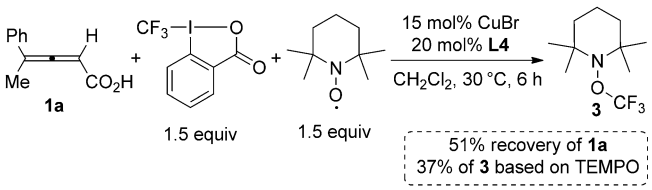


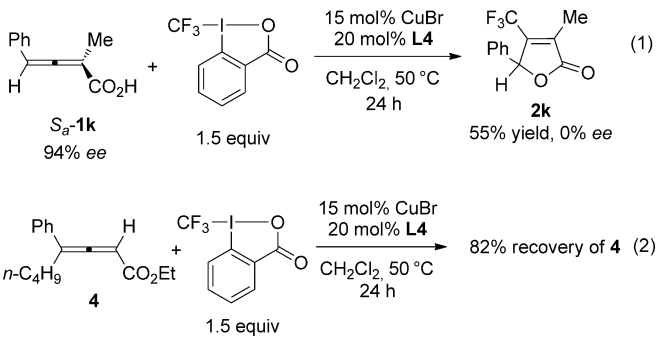
Figure 1. ORTEP representation of **2g** (50% probability).

To gain insight into the mechanism of this reaction, a radical scavenger 2,2,6,6-tetramethyl-1-piperdinyloxy (TEMPO) was added under the standard reaction conditions; there was no trifluoromethylation product **2a** formed and radical coupling TEMPO-CF_3 adduct **3** was formed in 37% yield (Scheme 3).



Scheme 3. Copper-catalyzed trifluoromethylation of **1a** in the presence of TEMPO.

We carried out more experiments to further probe some details of the proposed mechanisms (Scheme 4). First, we carried out the reaction with enantioenriched *Sa*-**1k** (94% *ee*) under standard conditions. The reaction delivered the corresponding product **2k** in 55% yield in its racemic form [Eq. (1), Scheme 4]; furthermore, when 2,3-allenoate **4** was subjected to the reaction conditions, no desired product was observed associated with 82% recovery of **4** [Eq. (2), Scheme 4], which demonstrates the importance of the copper carboxylate for the radical addition to allene unit mediated by its coordination with Cu. In addition, it was observed that when the phenyl group was replaced with *n*-hexyl group, no reaction occurred, indicting the importance of the benzylic stabilization effect for the radical addition reaction of trifluoromethyl radical.



Scheme 4. Attempted copper-catalyzed trifluoromethylation of *Sa*-**1k** and 2,3-allenoate **4**.

We may rule out the direct electrophilic trifluoromethylation of 2,3-allenoic acid (**1**) with Togni's II reagent since no product was obtained in the absence of copper complex (entry 12, Table 1). Based on these results, we propose the possible mechanisms for this reaction. Initially, coordination of the ligand with CuBr would generate catalytically active copper species **A**. The interaction of **A** with Togni's reagent II would generate a radical intermediate **B**, which would further generate CF_3 radical by releasing (2-iodobenzoyloxy)copper(II) bromide (**C**). In the presence of 2,3-allenoic acid **1**, **C** would generate corresponding coordination complex **D** by releasing of 2-iodobenzoic acid, which was indeed isolated from the reaction mixture as the byproduct. The coordinated allenyl moiety of intermediate **D** would be attacked by trifluoromethyl radical to form the π-allylic copper(III) intermediate **E**,[10,11] followed by reductive elimination to release the desired product **2** associated with the regeneration of the catalytically active Cu^I species **A** (Path A, Scheme 5). An alternative is the capture of in situ generated trifluoromethyl radical by the allene moiety of 2,3-allenoic acid **1** producing the corresponding allyl radical species **F**. Subsequently, this intermediate would undergo oxidation with Cu^{II} species **C** yielding allyl cation **G** with release of 2-iodobenzoic acid while regenerating copper(I) catalyst **A.** **G** would cyclize to deliver the product through intramolecular attack (Path B, Scheme 5).

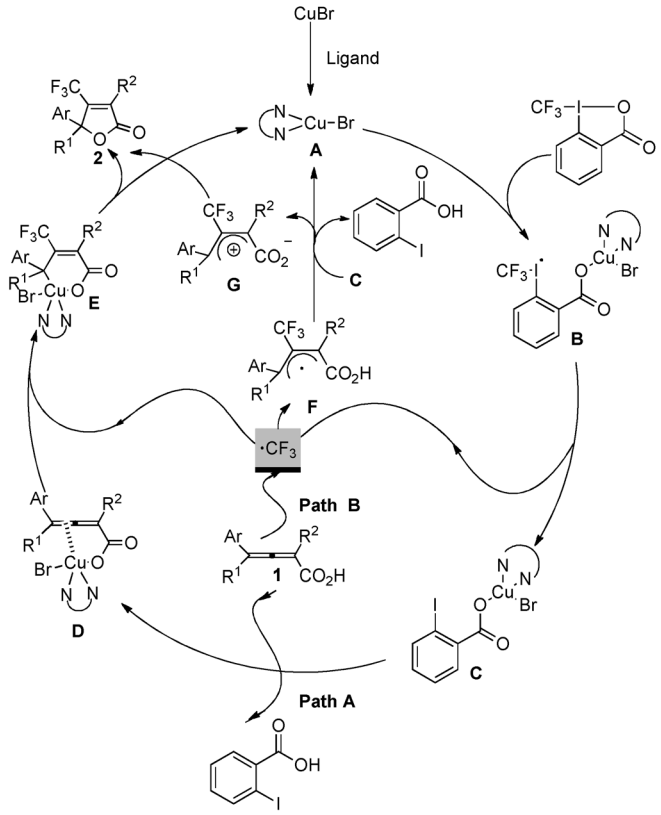
In summary, we have developed an efficient copper-catalyzed cyclic oxytrifluoromethylation of 2,3-allenoic acids under mild conditions, providing a straightforward way to β-trifluoromethylated butenolides. Due to the potentials of trifluoromethyl substituted butenolides, this method will be of interest to organic and medicinal chemists. Further studies in this area including the asymmetric vision of this reaction with chiral *N*,*N*-ligands are ongoing in our laboratory.

Experimental Section

Typical procedure for the synthesis of 5-methyl-5-phenyl-4-trifluoromethyl-2(5*H*)-furanone (2a): CuBr (22.3 mg, 0.15 mmol), **L4** (42.5 mg, 0.20 mmol), Togni's reagent II (474.5 mg, 1.5 mmol), **1a** (173.9 mg, 1.0 mmol), and 5 mL of CH_2Cl_2 under argon were added sequentially to

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Scheme 5. Plausible mechanisms of the reaction.

a flame-dried Schlenk tube. The resulting mixture was stirred at 30 °C. When the reaction was complete, as monitored by TLC, the crude reaction mixture was filtrated through a short pad of silica gel eluted with ether (20 mL). After evaporation, the residue was analyzed by NMR spectroscopy, and then purified by chromatography on silica gel (eluent: petroleum ether/ethyl acetate = 30:1) to afford **2a** (216.1 mg, 89 %) as an oil: ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.44–7.27 (m, 5H, Ar-H), 6.54 (s, 1H, =CH), 2.05 ppm (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 168.5, 159.8 (q, J = 34.4 Hz), 135.3, 129.3, 128.8, 125.3, 122.7 (q, J = 4.0 Hz), 120.3 (q, J = 270.6 Hz), 87.7, 23.1 ppm; ^{19}F NMR (282 MHz, $CDCl_3$): δ = −62.6 (s, 3F); IR (neat): $\tilde{\nu}$ = 2988, 2901, 1777, 1251, 1172, 1142, 1067 cm^{-1}; MS (EI) m/z (%) 242 (M^+, 42.97), 227 (100); HRMS calcd for $C_{12}H_9F_3O_2$ [M^+]: 242.0555; found: 242.0560.

Acknowledgements

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Keywords: allenoic acids · catalysis · copper · oxytrifluoromethylation · radicals

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[9] Crystal data for compound **2g**: $C_{12}H_8BrF_3O_2$, M_r = 321.09, orthorhombic, space group *Pccn*, final R indices [$I > 2\sigma(I)$], $R1$ = 0.0282, $wR2$ = 0.0648, R indices (all data) $R1$ = 0.0362, $wR2$ = 0.0684, a = 8.5817(6), b = 11.1375(8), c = 25.2045(17) Å, α = 90°, β = 90°, γ = 90°, V = 2409.0(3) Å^3, T = 173(2) K, Z = 8, reflections collected/unique

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26077/2136 (R_{int} = 0.0375), number of observations [$>2\sigma(I)$] 1827, parameters: 163. CCDC-931579 (**2g**) contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

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